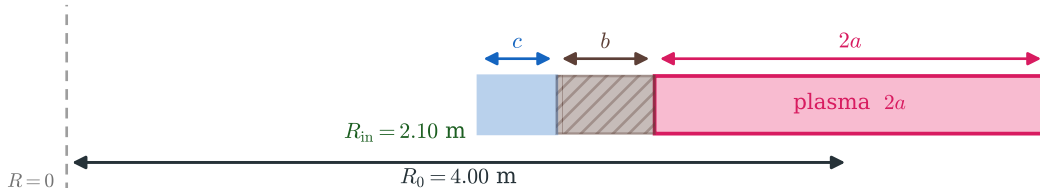


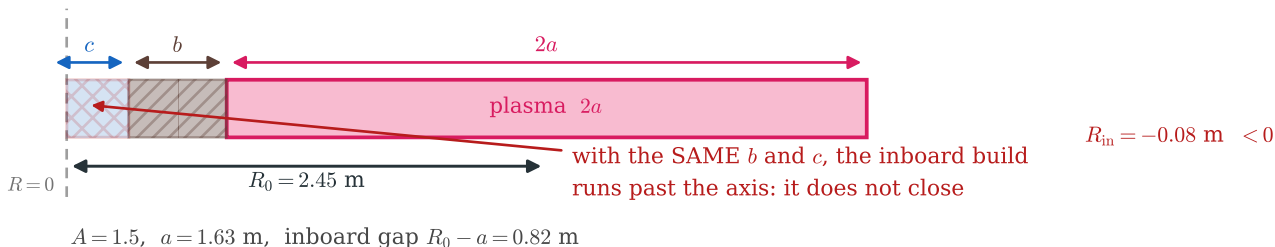
THE SAME INVENTORY, TWO ASPECT RATIOS — at fixed $R_0 a = C_{\text{wall}}$

conventional, $A \simeq 4$



$A = 4.0$, $a = 1.00 \text{ m}$, inboard gap $R_0 - a = 3.00 \text{ m}$

low aspect ratio, $A \simeq 1.5$



Both rows hold $R_0 a = 4.00 \text{ m}^2$, so the first-wall area is the same. $b = 0.50 \text{ m}$ and $c = 0.40 \text{ m}$ are fixed IN METRES: they are set by neutron attenuation and by conductor and structure, not by how round the plasma is — scaling them with a would assume the answer. Everything above is then arithmetic: $a = \sqrt{C_{\text{wall}}/A}$, $R_0 = Aa$.

NEITHER design is superior here, and the figure does not say one is. It says something narrower: low A cannot be adopted while holding b and c fixed — the inboard allocations have to be renegotiated. That is exactly what the following sections do, and why they quote their own shielding allocation and centre-column envelope instead of inheriting the conventional ones. The physics inventory has not changed; the room it fits into has.

Explanatory sketch; midplane cuts at a common radial scale. No HTS brand and no ST divertor: neither is derived yet.